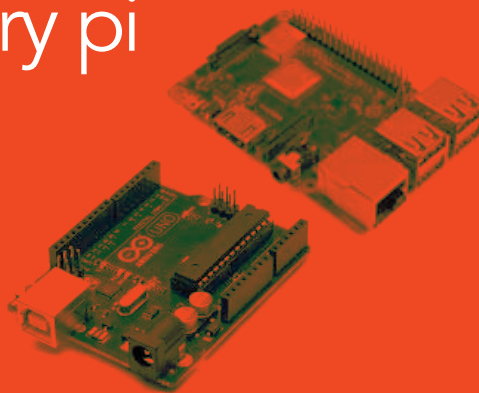


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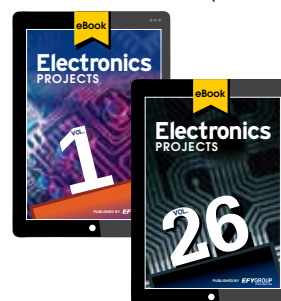
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and the receiver collects the reflected signal if the object is within the range.

### Software

Arduino program *dispenser.ino*, used for the prototype, should be uploaded to Arduino Uno. For this, open source software Arduino IDE is required.

In the program, Arduino's pins 3 and 13 are identified for digital output and pin 2 for digital input. Pin A0 is used for reading analogue signal. Main functions of the Arduino code are explained below.

**PinMode( ).** It configures the specified pin to behave either as an input or an output. See the description of digital pins for their functionality. They can also be used as PWM pins.

**DigitalWrite( ).** If the pin has been configured as an output with `pinMode( )`, its voltage will be set to the corresponding value: 5V (or 3.3V on 3.3V boards) for high, 0V (ground) for low.

**Serial.** It is used for communication between the Arduino board and a computer or other devices. All

Arduino boards have at least one serial port (also known as UART or USART), and some have several.

**begin( ).** It sets the data rate in bits per second (baud) for serial data transmission. For communicating with the computer, use one of these rates: 300, 600, 1200, 2400, 4800, 9600, 14400, 19200, 28800, 38400, 57600, or 115200. You can, however, specify other rates, for example, to communicate over pins 0 and 1 with a component that requires a particular baud rate.

**AnalogRead( ).** It reads the value from the specified analogue pin. Arduino boards contain a multichannel, 10-bit analogue-to-digital converter. This means that it will map input voltages between 0 and the operating voltage (5V or 3.3V) into integer values between 0 and 1023.

**println( ).** It prints data to the serial port as human-readable ASCII text followed by a carriage return character.

**Delay( ).** It is used to pause the program for the amount of time (in milliseconds) specified as parameter.

**delayMicroseconds( ).** It pauses the program for the amount of time (in microseconds) specified by the parameter. (There are a thousand microseconds in a millisecond and a million microseconds in a second.)

### Calculation of distance

The signal travels from transmitter to object and then returns to receiver. So, when you calculate distance of the object from sensor by multiplying time and speed, you have to divide it by two. The distance is calculated using the relationship:

$$\text{Distance} = (\text{Time} \times \text{Velocity}) / 2$$

Here, velocity of the wave is 340m/s, that is, 0.0340cm/microsecond.

$$\text{So, distance in centimetres} = (\text{Time} \times 0.0340) / 2$$

Currently, the largest value that will produce an accurate delay is 16,383 microseconds. This could change in future Arduino releases. For delays longer than a few thousand microseconds, the `delay( )` function can be used.